Stefan Milinovic

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Motivated software developer specializing in large-scale interactive educational games, automated testing frameworks, and robust software architectures. Skilled in problem-solving and delivering data-driven solutions, with a passion for software design, systems programming, and educational technology. Dedicated to continuous learning, mentoring, and applying clean engineering approaches to complex software systems.

TECHNICAL PROFICIENCIES

Languages: Java, Python, C, GDScript, JavaScript, TypeScript, SQL, Bash/Shell, HTML/CSS
Tools: Git, GitHub Actions, Linux, Godot 4, gdUnit4, SonarQube, Three.js, React, Node.js, Vite, AWS, Power BI

EDUCATION

BENG IN SOFTWARE ENGINEERING **GPA 3.80**
York University, **Toronto** *Expected graduation Apr 2028*

Relevant Coursework

- Advanced Object-Oriented Programming (EECS 2030, Grade: A+)
- Fundamentals of Data Structures (EECS 2101, Grade: A)
- Operating System Fundamentals (EECS 3221, Grade: A+)
- Introduction to Database Systems (EECS 3421, Grade: A)
- Digital Logic Design (EECS 3201, Grade: A+)

Extracurricular Activities

- YorkU Robotics Society: Member of YorkU Robotics Society, contributed to Rover Team's software subteam, developing and optimizing prototype code for competitions (CIRC, URC).

EXPERIENCE

SOFTWARE DEVELOPER CO-OP **Toronto, ON**
Ontario Science Centre *Jan - Present (Dec 2026)*

- Designed and developed large-scale interactive educational games and digital exhibits aimed at youth science education (~age 12), translating complex STEM topics into intuitive, gamified software systems.
- Architected core application systems using object-oriented principles, modular event buses, and design patterns (FSM, Strategy, Dependency Injection) across 57,000+ lines of production GDScript and JavaScript.
- Engineered an automated testing framework comprising 840 unit and integration tests across 75 test suites (gdUnit4), implementing mock actors, sandboxed filesystems, and headless CLI test execution.
- Authored 89 technical documentation deliverables, including 4 Architecture Decision Records (ADRs), contributor onboarding guides, and non-programmer data authoring specs to standardize team development.
- Configured CI/CD quality gates via GitHub Actions, SonarQube static analysis, and custom Python linting scripts to enforce cyclomatic complexity caps and translation schema integrity.
- Deployed interactive exhibits for 5,000+ visitors at the Harbourfront Centre, optimizing performance for 4K displays, multi-touch kiosk inputs, and WCAG 2.1 AA accessibility standards.

DATA ANALYST INTERNSHIP **Toronto, ON**
Havas CX Canada *May - Aug 2024*

- Conducted QA testing on iOS, Android, and web apps for clients, including Rogers Bank and Scene+, ensuring seamless functionality across platforms.
- Migrated legacy Universal Analytics implementations to GA4, debugging issues with Google Tag Manager and APIs to maintain accurate client data.

- Developed dashboards in Power BI, Tableau, and Data Studio, improving visibility into project costs and KPIs across departments.
- Delivered performance reports and SEO audits, strengthening decision-making and online visibility.

PROJECTS

- **Capto Carbon** *Action-Roguelike Systems Game – 2026*
  - Engineered a deterministic 5-pass procedural level generator using layered directed acyclic graphs (DAGs) and an independent structural rule-validation module for reproducible, bug-free map generation.
  - Programmed a modular enemy AI system using finite state machines and a tactical vantage-point planner that ring-samples candidate positions to verify line-of-sight and navmesh reachability before repositioning.
  - Built a crash-safe save-persistence system featuring atomic temp writes, versioned migrations, read-only safety locks against newer builds, and forced browser IndexedDB flushes for the web PWA build.
- **Launch Into Rockets** *3D Educational Web Experience – Spring 2026*
  - Built a scroll-driven 3D web application in Three.js and GSAP across 19 narrative stages, using arc-length polyline parameterization to project 3D GLTF vector curves into live, pointer-draggable 2D SVG overlays.
  - Engineered a reversible animation state machine on Three.js AnimationMixer and built a cross-browser transparent video streaming system using codec feature probing (VP9/HEVC).
- **AI Resume Screener** *ATS Matching Tool - Aug 2025*
  - Built a web app in React, TypeScript, and TailwindCSS that analyzes resumes against job postings, generating ATS scores (0–100) with feedback on skills, tone, structure, and content.
  - Integrated OpenAI API via Puter.js for serverless AI, authentication, and file handling, enabling secure uploads and real-time improvement tips for stronger job alignment.
- **Website Redesign** *E-commerce Client - July 2025*
  - Redesigned an e-commerce site frontend using WordPress, React, and CSS, modernizing product, cart, and checkout pages, and fixing broken payment options (Google Pay, Apple Pay, PayPal, Visa).
  - Improved performance metrics with ~50% more organic traffic (100 visitors/month) and ~20% longer sessions, alongside stronger SEO visibility post-launch.